

User defined functions

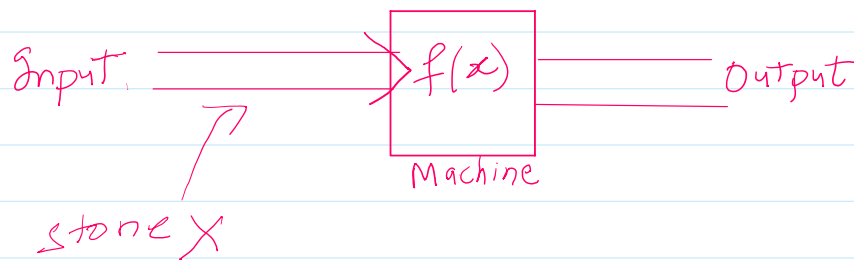
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Functions which are present in Python is called system defined function. We need to call the function and it will do its specific job.

What is a function?

A function is a group of code, which solves a specific problem.

⇒ Mixer grinder ()
⇒ juicer ()
⇒ chopper ()
⇒ stove (electric, gas)



User-Defined Functions in Python

A user defined function in Python is a block of organized, reusable code that you create to perform a specific task. You can call (run) your function whenever you need that task done instead of repeating code.

Why use Functions?

- **Reduce repetition:** Write code once, use it many times.
- **Organize and clarify code:** Functions break complex problem into smaller manageable pieces.
- **Improve readability:** Named function describe what they do, making code easier to understand.

Defining a function

Use the def key word, a function name, parenthesis (With parameters if needed) And a colon. The function body is intended:

starts the function definition.

```
def say_hello():  
    print("Hello!")
```

function name holds the parameter (if any) The indented code inside is the function body.

For example:

```
def greet():  
    print('Welcome to Python!')
```

```
#main code  
greet()  
greet()  
greet()  
greet()
```

Output:

```
Welcome to Python!  
Welcome to Python!  
Welcome to Python!  
Welcome to Python!
```

Function with Arguments

```
def greet(name:str):  
    print(f'Welcome {name} to Python!')
```

```
#main code  
greet('Vannya')  
greet('Vania')  
greet('Vaanya')  
greet('Vanya')
```

Output:

```
Welcome Vannya to Python!  
Welcome Vania to Python!  
Welcome Vaanya to Python!  
Welcome Vanya to Python!
```

Function return values by using the return statement. The job of return statement is, it takes the execution control along with the value back to the calling statement.

'return' statement has three forms.

- i. 'return' : this takes the execution control back to the calling line and execution control moves to the next statement.
- ii. 'return value': the value is also returned with the execution control on the calling line, and execution moves further.
- iii. 'return <condition>': this always returns 'boolean' value.

```
def add(a:float, b:float):  
    return a+b  
def subtract(a:float, b:float):  
    return a-b  
def product(a:float, b:float):  
    return a*b  
def division(a:float, b:float):  
    return a/b
```

```

def quotient(a:float, b:float):
    return a//b
def remainder(a:float, b:float):
    return (a%b)
def power(a:float, b:float):
    return (a**b)
#main code
print('='*50)
print('Welcome to Simple calculator')
print('='*50)
while True:
    print('Press +, -, *, /, //, %, **, x(exit): ')
    op = input()
    match op:
        case '+':
            n1 = float(input('Enter first number: '))
            n2 = float(input('Enter second number: '))
            res = add(n1, n2)
            print(f'Result: {res}')
        case '-':
            n1 = float(input('Enter first number: '))
            n2 = float(input('Enter second number: '))
            res = subtract(n1, n2)
            print(f'Result: {res}')
        case '*':
            n1 = float(input('Enter first number: '))
            n2 = float(input('Enter second number: '))
            res = product(n1, n2)
            print(f'Result: {res}')
        case '/':
            n1 = float(input('Enter first number: '))
            n2 = float(input('Enter second number: '))
            res = division(n1, n2)
            print(f'Result: {res}')
        case '//':
            n1 = float(input('Enter first number: '))
            n2 = float(input('Enter second number: '))
            res = quotient(n1, n2)
            print(f'Result: {res}')
        case '%':
            n1 = float(input('Enter first number: '))
            n2 = float(input('Enter second number: '))
            res = remainder(n1, n2)
            print(f'Result: {res}')
        case '**':
            n1 = float(input('Enter first number: '))
            n2 = float(input('Enter second number: '))
            res = power(n1, n2)
            print(f'Result: {res}')
        case 'x':
            print('Good Bye')
            exit(0) #terminate the infinite loop.
        case _:

```

```
print('Wrong option selected')
```

WAF which checks the number is even or odd, if the number is even it returns 'True' otherwise 'False'

WAP which keep asking for a number and computer is detecting if the number is even or odd, press 'x' to exit.

Home work

1. Temperature Converter

Write a function that takes a temperature in Celsius as an argument and returns the corresponding temperature in Fahrenheit. Use the formula ($F = C \times 9/5 + 32$). Prompt the user to enter a Celsius temperature and display the result.

2. Largest of Three Numbers

Create a function that accepts three numbers as parameters and returns the largest one. Test your function with various inputs and show the result for each set.

3. Palindrome Checker

Define a function that takes a string as input and returns True if the string is a palindrome (the same forward and backward), and False otherwise. Ignore case and spaces while checking for palindromes.

4. Summing a List

Write a function that accepts a list of numbers and returns their sum. Then, ask the user for a set of numbers, convert them into a list, and use your function to compute the total.

5. Count Vowels

Create a function that takes a string and returns the number of vowels (a, e, i, o, u) in it. Make your function case-insensitive.

6. Factorial Calculator

Write a function to compute the factorial of a positive integer (n!). Do not use Python's built-in `math.factorial`. Use a for loop inside your function.

7. Prime Number Checker

Define a function that determines if a given number is prime. The function should return True for prime numbers and False otherwise. Test your function with several numbers.

8. Custom Greeting

Write a function that takes a name and a time of day (like "morning", "afternoon", or "evening") as arguments and prints a personalized greeting such as, "Good morning, Alex!"